

tie their show lace or ask someone else to do it for them. The importance of this becomes even greater when it comes to those who have lost their hands; they won't have to depend on anyone else or be restricted to specific types of footwear.

The fun part is that this project was actually inspired by the movie *Back to The Future II*! That's right, now you can make your own Marty McFly self-lacing shoes.

The project is a simple one and is based on the Arduino microcontroller. The way it functions is delightfully simple. When you step into your shoe, a force sensor senses the pressure and in turn activates two servo motors which apply tension to the laces to tighten the shoe. A touch switch reverses the servo (you can modify this part if you can think of a way which is more convenient to you).

A hightop shoe with a lot of padding, Arduino microcontroller, Motor shield (from Adafruit), Force sensor (from Adafruit), Servo Motors (from adafruit), Sheet metal, LED and a couple of resistors, 9 Volt case with built-in battery clip and switch, insulated copper wire, plastic zip ties, plastic 1/2" cable loops for organising cables and 1/8" braided nylon paracord. All these materials are easily available online or off the shelf.

Although this may look big and clumsy, the maker has gone ahead and created a much smaller version for commercial purposes.

Make it yourself:

<http://dgit.in/PowerLace>

Turn signal biking jacket: Making city cycling easier

One of the main issues faced by those who cycle in the city are giving turn signals. The dangers of this are losing your balance when sticking your hand out to signal a turn or your hand being accidentally hit by a vehicle or several other possibilities.

Now this problem can be solved easily by buying an electronic indicator kit for the cycle. Then again, this has problems with theft and damage and that a good kit can get expensive (let alone expensive to replace). In comes turn signal biking jackets to the rescue. Now you can display turn indicators on the backside of your jacket with LED lights and keep your hand safe and also let people know where you are headed. It uses conductive thread and sewable electronics so that it is comfortable and washable as well.